

Spectrum of Logic

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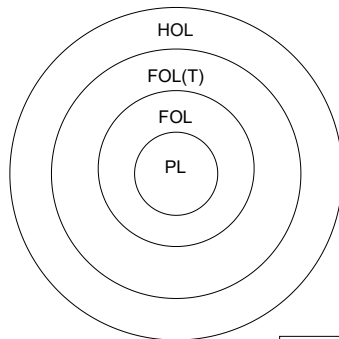
Theorem Proving

- Spectrum of Logic
- Decision Problems

Simpler the logic – Easier will be the decidable test

- Propositional Logic (PL)
- First Order Logic (FOL)
- First Order Logic with Theories(FOL(T))
- Higher Order Logic (HOL)

Logic Hierarchy



PL - Propositional Logic
FOL - First Order Logic
FOL(T) - First Order Logic with Theories
HOL - Higher Order Logic

Decision Problems

- Satisfiability
- Validity
- Contradiction / Unsatisfiable

Decision Problems

Formula(F)

Interpretation of Formula : Instantiation of the variables in F from domain D. Eg:

$$F = x1 \vee x2 \vee x3$$

$$I = x1 \vdash 1, x2 \vdash 0, x3 \vdash 0$$

Satisfiability

Whether Formula(F) is true under an Instantiation(I) of F?

$$I \models F? - \textit{satisfiability}$$

It is an NP complete problem for propositional logic.

Decision Problems

Validity

Formula(F) is valid if it is satisfiable for all interpretations

$$\vdash F$$

Contradiction

Formula(F) is contradiction if it is not satisfiable.

Propositional Logic(PL)

PL

PL consists of propositional variables and boolean connectives.

- Modelling Eg 1:
 1. If x is a prime greater than 2 then x is odd.
 2. If it is not the case that x is not a prime greater than 2.
 3. x is not odd

Check whether all 3 statements are true together?

Propositional Logic(PL)

Step 1 : Labelling

Mark part of statements as propositional variables.

i.e a : x is prime greater than 2.

b : x is odd.

Given three statements after labelling are as follows:

$$F1 : a \rightarrow b$$

$$F2 : \neg \neg a$$

$$F3 : \neg b$$

is $F1 \wedge F2 \wedge F3$ satisfiable?

Two proof methods exist:

1. Proof by Deduction
2. Proof by Enumeration

Proof by Deduction : We have set of axioms/rules using which we prove.

- Contradiction :
$$\frac{Q \quad \neg Q}{F}$$

- Double Negation :
$$\frac{\neg\neg Q}{Q}$$

- Modus Ponens(MP) :
$$\frac{Q1 \rightarrow Q2 \quad Q1}{Q2}$$

Proof By Deduction

Proof of the above example 1

1. $a \rightarrow b$ (*Premises*)
2. $\neg\neg a$
3. a (Double negation)
4. b (1, 3 and Modus Ponens)
5. $\neg b$ (*Premises*)
6. False (Contradiction)

Proof by Enumeration

Construct the truth table for Formula(F).

First Order Logic (FOL)

FOL

consists of PL , predicate constants and functions.

Predicates

Boolean functions that return 1/0.

$H(x)$: x is human.

$M(x)$: x is mortal.

First Order Logic (FOL)

Eg 1.

1. Human are mortal.
2. Monu is Human.
3. Monu is mortal.

The formulas will be as follows:

F1 :

$$\forall x(H(x) \rightarrow M(x))$$

F2 : $H(\text{Monu})$

F3 : $M(\text{Monu})$

First Order Logic with Theories (FOL(T))

Eg 2.

There are arbitrarily large prime numbers.

i.e For a prime number there always exist a larger prime number that is prime.

$$\forall x \exists p (p > x) \wedge (\forall v1 \forall v2 (v1 > 1 \wedge v2 > 2 \rightarrow p \neq v1.v2))$$

Functions: $\text{greater}(x, y)$ and $\text{mult}(x, y)$

Theory v/s Example

Theory	Example
Propositional Logic	$x_1 \wedge \neg x_2 \wedge \neg x_3$
FOL/Predicate Logic	$\forall x (H(x) \rightarrow M(x))$
Equality	$(y_1 = y_2) \vee (y_2 = \neg y_3)$
Linear Arithmetic	$(2x_1 + x_3 > 3) \vee (x_4 + x_5 > 6)$
Bit Vectors	$(a \gg 2) \wedge c$
Arrays	$(i = j) \wedge (a[i] = 1 \rightarrow a[i] = 1)$
Pointers	$p = 2 \wedge *p = 5 \rightarrow *q = 5$
Combined	all of the above

Higher Order Logic (HOL)

HOL

Consists of FOL, predicate variables and function variable.

$$\forall p \forall x [p(x) \vee \neg p(x)]$$

Here for all possible interpretations of predicates the formula will be checked.

Higher Order Logic (HOL)

$$\exists F \{ F(a) = b \wedge \forall x \{ p(x) \rightarrow (F(x) = g(x, F(f(x)))) \} \}$$

Interpretation:

- Domain D
- Assign one element of D to a , b .
- An unary function

$$f : D \rightarrow D$$

- A binary function to

$$g : D \times D \rightarrow D$$

- An unary predicate

$$p : D \times \{T, F\}$$

- For quantified functions, variable, assignment is not required.

Higher Order Logic (HOL)

$$\exists F \{ F(a) = b \wedge \forall x \{ p(x) \rightarrow (F(x) = g(x, F(f(x)))) \} \}$$

Interpretation:

- D: Set of natural numbers
- $a = 0$, $b = 1$
- $f(x)$ is the predecessor function.

$$f(x) = x - 1$$

$$f(0) = 0$$

- g is product function

$$g(x, y) = x \times y$$

- P is checking x greater than 0.

$$p(x) : x > 0$$

- For quantified functions, variable, assignment is not required.

Higher Order Logic (HOL)

$$\exists F \{ F(a) = b \wedge \forall x \{ p(x) \rightarrow (F(x) = g(x, F(f(x)))) \} \}$$

- There exists a function F over N such that $F(0) = 1$ and for every $x \in N$, if $x > 0$, then $F(x) = x \times F(x - 1)$.
- This is clearly a factorial function.
- There can be many other interpretation of this formula.

Thank You